Total Printed Pages: 02

Roll No. :.....

PYB-101

Examination –May- 2024

B.Tech. I/II Sem : Common to all Branches

Applied Physics

Time : 2 Hrs

Max. Marks : 60
Min. Marks : 19

Note: Total number of questions are 05. All Questions are compulsory. Each Question has 4 parts (a, b, c, d). Part a & b are compulsory while Part c & d has internal Choice. Assume missing data, if any.

Word limit be observed as follows:
Part a & b – Objective type/Fill in the blanks/One sentence answer.
Part c – Max 100 words and Part d – Max 400 words.
Word limit NOT to be followed for diagram, numerical, derivation.

				Marks	BL	CO
Q.1	(a)	Particles can interfere just like light does.	True / False	02	2	1
	(b)	Energy is	Discrete / Continuous	02	2	1
	(c)	Deduce the energy operator.		03	3	1
		OR				
		Explain de Broglie relation.		03	3	1
	(d)	Explain Compton scattering.		05	4	1
		OR				
		Derive time dependent Schodinger wave equation.		05	4	1
Q.2	(a)	Laser light is.....	coherent / incoherent.	02	2	2
	(b)	When there are more atoms in an excited state than in the lower energy state, it is called population	excitation / inversion	02	1	2
	(c)	Explain numerical aperture of an optical fiber.		03	3	2
		OR				
		Discuss various loss mechanisms in optical fiber.		03	3	2
	(d)	Explain working of a Ruby laser .		05	4	2
		OR				
		Explain holography and its applications (any two) .		05	4	2
Q.3	(a)	The gap between valence and conduction band in a semiconductor is called		02	1	3
	(b)	The resistance of semiconductors is higher than conductors.		02	2	3

			True / False		
	(c)	Differentiate between direct and indirect band gap semiconductors.	03	3	3
		OR			
		Explain significance of Fermi energy level.	03	4	3
	(d)	Discuss construction and working of LED.	05	4	3
		OR			
		Discuss construction and working of solar cell.	05	4	3
Q.4	(a)	Superconductors are sensitive to magnetic field True / False	02	2	4
	(b)	Superconductors have infinite electrical conductivity True / False	02	2	4
	(c)	Write a note on carbon nanotubes	03	3	4
		OR			
		Explain transition temperature in superconductors.	03	3	4
	(d)	Explain Meissner effect in superconductors.	05	4	4
		OR			
		Discuss few applications of nanotechnology.	05	4	4
Q.5	(a)	If a dielectric possesses permanent dipoles, it is classified as..... polar / nonpolar	02	2	5
	(b)	The capacitance of a capacitor increases by inserting a dielectric True / False	02	2	5
	(c)	Discuss Gauss law in dielectrics.	03	3	5
		OR			
		Explain piezoelectric effect.	03	3	5
	(d)	Deduce the relation between E, D and P vector in a dielectric.	05	3	5
		OR			
		Discuss any three applications of piezoelectric effect.	05	3	5

Bloom's Taxonomy Levels (BL): 1-Remembering, 2- Understanding, 3- Applying, 4- Analyzing, 5- Evaluating, 6-Creating

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